

Digital and analog quantum algorithms for the nuclear many-body problem

Thesis of Samuel Aychet-Claisse
Summary and discussion

Morten Hjorth-Jensen

University of Oslo
Defense at Université Paris-Saclay

1 October 2026

The nuclear many-body problem and quantum computing

- Correlations and symmetry constraints make nuclear ground states and spectra difficult to compute accurately.
- Configuration interaction, coupled-cluster, Green's-function and Monte Carlo methods each control different approximations.
- Quantum algorithms offer alternative representations and measurements of correlated states.
- Their scientific value depends on the **total cost at a specified accuracy**: preparation, optimization, measurement and error control.

This thesis tests that proposition on controlled nuclear models, using both digital algorithms and analog controls.

Thesis, Chs. 1–2 and pp. 115–117. Context: Ayral et al., Eur. Phys. J. A 59 (2023).

Two benchmarks with different purposes

Pairing: correlations and particle-number conservation

$$H_P = \sum_{k=1}^D \epsilon_k (n_k + n_{\bar{k}}) - g \sum_{k \neq l} P_k^\dagger P_l, \quad P_k^\dagger = a_k^\dagger a_{\bar{k}}^\dagger.$$

- Main digital benchmark: $A = 8$ particles in $D = 8$ doubly degenerate levels.
- Seniority-zero space: $\binom{8}{4} = 70$ states, compared with $\binom{16}{8} = 12870$ in the full fixed- A space.

Deuteron: encoding and hardware constraints

- Leading-order pionless-EFT model in a truncated oscillator basis.
- Three basis states provide a small, exactly checkable analog target.

Thesis, pp. 22–26, Eqs. (1.24)–(1.31). Earlier digital deuteron calculation: Dumitrescu et al., PRL 120, 210501 (2018).

The thesis connects algorithms to measurable observables

Contribution	Main result	Evidence
Adaptive states	Four ADAPT variants for pairing	Ideal digital emulation
Green's functions	Subspace reconstruction and measurement budget	Ideal emulation and finite-shot sampling
Analog protocols	Controls and encoding tailored to neutral atoms	Constrained pulse simulations

The analog work delivers a protocol ready for a hardware test.

The thesis does not report a completed Ruby experiment or demonstrated quantum advantage.

Thesis, Chs. 3–6, especially pp. 114–117.

Adaptive state preparation: pool and selection both matter

$$|\Psi_n\rangle = R_{i_n}(\theta_n) \cdots R_{i_1}(\theta_1) |\Psi_{\text{HF}}\rangle.$$

Variant	Operator pool	Selection
ADAPT-St	Pair-exchange rotations	Largest gradient
ADAPT-Min	Pair-exchange rotations	Lowest attainable energy
ADAPT-Min2	Neighboring reflection blocks	Lowest attainable energy
ADAPT-Fix	Neighboring reflection blocks	Predetermined order

For the pools studied, the one-parameter energy has a closed form:

$$E(\theta) = C_0 + C_1 \cos(\theta + \gamma_1) + C_2 \cos(2\theta + \gamma_2).$$

Five ideal energy evaluations determine the curve used by ADAPT-Min.

Thesis, pp. 53–59. ADAPT context: Grimsley et al., Nature Communications 10, 3007 (2019).

Pairing benchmarks support compact, accurate ansätze

$$\delta E = \frac{E - E_{\text{exact}}}{E_{\text{HF}} - E_{\text{exact}}} \quad (\text{error relative to the correlation energy}).$$

- For $A = D = 8$, successful variants reach $\delta E < 1\%$ with about 20 operators in the reported convergence tests.
- ADAPT-Min improves the early iterations. ADAPT-St catches up around 15 iterations in the examples.
- At 20 iterations, the successful variants generally improve on projected BCS across the crossover region.
- Projected BCS remains competitive at strong coupling and uses only D variational parameters.

Thesis, pp. 59–63, Figs. 3.5–3.7. The BCS critical coupling is $g_c/\Delta E \simeq 0.298$ for this benchmark.

Compact circuits come with an optimization cost

- **Re-optimization is decisive.** In the $A = D = 6$, $g/\Delta E = 0.6$ test, it improves the energy error by orders of magnitude without adding operators.
- **Greedy selection can stall.** ADAPT-Min2 exhibits a two-step reflection loop in the $g/\Delta E = 0.6$ example.
- **Fewer iterations need not mean fewer measurements.** The reported selection cost is $\frac{5}{2}D(D-1)$ basis evaluations per step for ADAPT-St and $\frac{15}{2}D(D-1)$ for ADAPT-Min.
- The dependence of the required iteration count on D at fixed accuracy remains unresolved.

Thesis, Table 3.1, pp. 58–63. A basis evaluation itself requires repeated shots.

Green's functions connect ground states to spectroscopy

The one-body propagator describes particle addition and removal. Its trace has the spectral representation

$$F(\omega) = \sum_{\nu} \frac{S_{\nu}^{+}}{\omega - \epsilon_{\nu}^{+} + i\eta} + \sum_{\mu} \frac{S_{\mu}^{-}}{\omega - \epsilon_{\mu}^{-} - i\eta},$$

$$\epsilon_{\nu}^{+} = E_{\nu}^{A+1} - E_0^A, \quad \epsilon_{\mu}^{-} = E_0^A - E_{\mu}^{A-1}.$$

- Pole positions give separation energies. Residues give addition and removal strengths.
- Spectral moments test more than the ground-state energy alone.
- This connects quantum algorithms with configuration mixing and Green's-function methods in nuclear structure.

Thesis, pp. 20–22, 65–68, 75. η is a spectral regulator, not a calculated decay width.

Quantum subspace expansion divides the work

Quantum measurements on one approximate even- A state

$$|\phi_{\alpha}^{\pm}\rangle = \Lambda_{\alpha}^{\pm}|\tilde{\Psi}_0^A\rangle, \quad O_{\alpha\beta}^{\pm} = \langle\phi_{\alpha}^{\pm}|\phi_{\beta}^{\pm}\rangle, \quad H_{\alpha\beta}^{\pm} = \langle\phi_{\alpha}^{\pm}|H|\phi_{\beta}^{\pm}\rangle.$$

Classical diagonalization and spectral reconstruction

$$H^{\pm}c_{\nu}^{\pm} = \tilde{E}_{\nu}^{A\pm 1}O^{\pm}c_{\nu}^{\pm}.$$

- The thesis uses $\Lambda_{\alpha}^{+} = a_{\alpha}^{\dagger}$ and $\Lambda_{\alpha}^{-} = a_{\alpha}$.
- Overlap measurements also supply the spectroscopic amplitudes.
- The expansion approximates the odd-system subspaces. Its operator choice controls the spectral information retained.

Thesis, pp. 67–69, Eqs. (4.4)–(4.10). Aychet-Claisse et al., PRC 113, 044324 (2026).

Pairing symmetries substantially reduce measurement cost

Quantity	Pairing result
Generic overlap and Hamiltonian entries ($\Omega = 2D$)	$4\Omega^2 = 16D^2$
After blocked-level and time-reversal simplifications	$4D$
Measurement bases with entangling basis rotations	$2D + 1$
Shots per basis for about 1% moment uncertainty	$\sim 10^4$

For $D = 8$: $17 \times 10^4 \simeq 1.7 \times 10^5$ shots per reconstruction,
excluding the ground-state optimization budget.

The shot estimate is empirical for small, noiseless systems. Quadratic total scaling with D is a tentative extrapolation.

Thesis, pp. 69–72 and 77–78, Fig. 4.5. Shot benchmark: $A = D = 8$, $g/\Delta E = 0.6$.

The propagator also gives access to neighboring odd systems

- The spectral-moment benchmarks favor ADAPT-Min over VAP-BCS and PAV-BCS at the tested couplings.
- VAP-BCS and ADAPT-Min reproduce the first four moments to within a few percent in the reported comparisons.
- Odd-system energies follow from the even-state calculation and reproduce odd–even staggering well.
- Remaining errors include the input state, the restricted subspace and finite sampling. Accurate energy differences can partly benefit from error cancellation.

Broader relevance: this construction also suggests a classical configuration-mixing route to odd nuclei.

Thesis, pp. 73–79, Figs. 4.2–4.5. Published study: Aychet-Claissé et al., arXiv:2509.02272 / PRC 113, 044324.

Neutral-atom controls change the algorithmic problem

An Ising-mode resource Hamiltonian has the form

$$H_R(t) = \frac{\hbar\Omega(t)}{2} \sum_i [\cos \varphi X_i - \sin \varphi Y_i] - \hbar\delta(t) \sum_i n_i + \sum_{i<j} \frac{C_6}{r_{ij}^6} n_i n_j.$$

- The atom interaction continues during state preparation and measurement rotations.
- Global controls constrain accessible states and measurable observables.
- Geometry links the couplings. Distances cannot be optimized independently.
- The thesis distinguishes a prospective XY device with local detunings from the Ruby configuration considered for the deuteron protocol.

Thesis, Ch. 5 and pp. 93–95, 103–114. Hardware capabilities here refer to the thesis's stated configuration.

The XY study exposes a reachability problem

- Fixed Hamming weight K preserves particle or pair number.
- $K = 1$ succeeds in the tested deuteron cases, but explores only an N -dimensional space.
- For $2 \leq K \leq N - 2$, local minima remain. Exact reachability is unresolved.

Pairing error δE in percent

(D, N_p)	Chain	Free plane
(4, 2)	2.4	0.13
(5, 2)	4.5	0.20
(6, 2)	5.6	0.75
(6, 3)	9.2	0.67

Best of the two initialization columns for each geometry in Table 6.1.

The one-pulse ansatz has $3N - 4$ adjustable parameters, compared with $\binom{N}{K} - 1$ for a general normalized real state.

Thesis, pp. 96–102, Table 6.1. N_p counts pairs, $g/\Delta E = 0.6$, evolution time 200 ns. Percentages converted from the table.

An encoding tailored to accessible states and measurements

The three deuteron basis states fit in the symmetric two-qubit space:

$$\Lambda|100\rangle = (|01\rangle + |10\rangle)/\sqrt{2},$$

$$\Lambda|010\rangle = (|00\rangle + |11\rangle)/\sqrt{2},$$

$$\Lambda|001\rangle = (|00\rangle - |11\rangle)/\sqrt{2}.$$

- The encoded Hamiltonian contains only II , $ZI + IZ$, $XI + IX$, ZZ , XX and YY terms.
- Three global measurement settings suffice in the ideal-rotation limit.
- The encoding addresses both state accessibility and observability. Reducing qubit count alone would not settle either issue.

Thesis, pp. 103–106, Eqs. (6.19)–(6.25).

Accurate preparation still leaves a measurement bias

Three-level deuteron calculation	Energy (MeV)	Error
Exact diagonalization of the target	-2.046	—
Reconstructed energy, ideal pulse response	-1.876	8.3%
Reconstructed energy, modulated response	-1.866	9.1%

- Ideal simulated preparation reaches about 10^{-12} relative energy error before measurement rotations.
- Always-on interactions during the rotations produce the dominant reconstruction bias in this test.
- Statistical uncertainty follows $\sigma_E/|E_{\text{exact}}| = 2.77/\sqrt{N_{\text{shots}}}$ per-setting shot count. More shots do not remove this bias.

Thesis, pp. 110–114, Table 6.2 and Eq. (6.43). Full hardware noise and a completed device experiment remain outside these results.

Scientific assessment and the next benchmark

The main contribution

A coherent treatment of state construction, spectral observables and hardware-constrained measurement, with explicit resource estimates and limitations.

What would strengthen the case for larger nuclear applications?

- Fixed-accuracy scaling against strong classical baselines.
- Separate uncertainty budgets for the nuclear model, ansatz, subspace, sampling and device controls.
- Richer interactions and particle-addition/removal spaces beyond pure pairing.
- A reproducible hardware demonstration that includes its measurement and optimization costs.

Assessment informed by the supplied opponent report, pp. 1–2, and thesis conclusions, pp. 115–117.

Discussion: scientific contribution and benchmarks

1. Your principal contribution

Which result do you regard as your most original contribution, and which part of it do you expect to remain useful as quantum hardware improves?

2. The role of the pairing model

Which favorable properties of pairing are essential to your results? What changes first when a term breaks seniority conservation?

3. A fair classical comparison

How would you test the scaling of ADAPT against VAP-BCS and an exact pairing solver at fixed accuracy, including optimization and measurement costs?

Discussion anchors: thesis, pp. 23–25, 59–63, 69–72, 115–116.

Discussion: adaptive algorithms and error measures

4. Greedy selection and long-term convergence

Why can ADAPT-St catch up with ADAPT-Min despite making smaller early energy gains? How would you compare them at equal shot budgets?

5. The reflection-loop failure

The $B(\theta)$ blocks are unitary but cannot become the identity. How does that permit the observed two-step loop, and what modification would prevent it?

6. Convergence with finite measurements

How would you combine correlation-energy error, energy variance and gradient uncertainty in a stopping rule when the exact solution is unavailable?

Discussion anchors: thesis, pp. 55–63, especially Figs. 3.5 and 3.7.

Discussion: Green's functions and uncertainty

7. The limits of the one-body subspace

If the input ground state were exact, which spectral features could still be missing with only $a^\dagger$ and a operators? What would adding $2p-1h$ and $2h-1p$ operators recover?

8. Checks beyond low-order moments

Which sum rules and positivity checks would you impose on the reconstructed spectrum? Can accurate low-order moments conceal incorrect individual poles or strengths?

9. Noisy overlap matrices

The shot study discards samples with non-positive overlap matrices. How frequent are these failures, and how would regularization change the bias, confidence intervals and shot budget?

Discussion anchors: thesis, pp. 67–78, including footnote 5 on p. 77.

Discussion: analog control and a decisive next experiment

10. Reachability versus optimization failure

In the XY pairing calculation, how would you determine whether the target lies outside the accessible family or whether the optimizer misses it? Would additional pulse segments resolve the distinction?

11. Measurement rotations as part of the experiment

How would you reduce or calibrate the 8–9% energy bias? Compare shorter rotations, extrapolation and treating the interacting pulse as a calibrated measurement.

12. The next nuclear demonstration

What system and observable would you choose next, and what accuracy, total runtime and classical baseline would make the result scientifically convincing?

Discussion anchors: thesis, pp. 99–114 and 116–117.